

claude-windows / a9fd4426-f234-4e4a-bfb4-1f2c8255c661

Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-muneem\a9fd4426-f234-4e4a-bfb4-1f2c8255c661\subagents\workflows\wf_076beba8-59c\agent-a21879c41090608c9.jsonl`  
- SHA-256: `fdb378ca030e427947439db20bbda8e3587864421e281b1a7547a7f3d88c0683`  
- Source modified: `2026-05-31T04:23:18+00:00`  
- Imported at: `2026-07-05T16:48:20+00:00`  
- project: `wf_076beba8-59c`  
- session_id: `a9fd4426-f234-4e4a-bfb4-1f2c8255c661`
```

Transcript

1. user (2026-05-31T04:22:16.489Z)

Adversarial Claim Verifier (voter 3/3)

Be SKEPTICAL. Try to REFUTE this claim. ?2/3 refutations kill it.

Research question

Identify the best LOCAL (self-hosted, single NVIDIA CUDA GPU) vision / document-AI / OCR models ? and, as a clearly-flagged approval-gated fallback only, paid cloud API services ? for parsing BANK STATEMENT PDFs into structured transaction tables (date, narration/description, withdrawal/debit, deposit/credit, running balance), for a LOCAL-FIRST personal-finance tool ("muneem").

Target documents: Indian bank statements (HDFC, Axis, ICICI, AU Small Finance) ? dense, multi-column, often SEMI-RULED or borderless tables, sometimes multiple accounts per statement; both born-digital (text layer present) and the harder cases where pdfplumber's extract_tables/coordinate clustering is unreliable (e.g. Axis, whose column layout varies between statements).

HARD CONSTRAINTS (a model that violates these is unusable for us):

1. MUST run 100% LOCALLY for statement contents (highly sensitive). NO cloud APIs by default. Paid/cloud is acceptable ONLY as an explicitly-approved fallback ? still report the best cloud options but label them clearly as "cloud/sensitive ? approval-gated, not default".
2. MUST be usable from Python (transformers / vLLM / Ollama / ONNX Runtime / native lib).
3. License MUST be permissive & commercially safe for BOTH the CODE and the MODEL WEIGHTS: MIT, BSD, Apache-2.0, ISC, HPND. AVOID and explicitly FLAG anything GPL/LGPL/AGPL, CC-BY-NC / non-commercial, research-only, or custom "community"/acceptable-use licenses with restrictions. This is critical ? many document-AI models and their training datasets are non-commercial (e.g. LayoutLMv3 weights are CC-BY-NC; Surya and Marker use GPL-or-commercial dual licensing; several VLM weights ship custom restrictive licenses). Verify the WEIGHTS license, not just the repo license.
4. Hardware: a single consumer/prosumer NVIDIA GPU. Give VRAM tiers (~8-12 GB consumer, ~16-24 GB) and which models/quantizations fit each.

Evaluate and compare these LOCAL approaches ? for each give: code license, MODEL-WEIGHTS license (state explicitly), latest release / maintenance status as of 2025-2026, VRAM need + quantization options, Python integration path, and reliability specifically on DENSE FINANCIAL TABLES / multi-column statements:

- Open vision-language models (VLMs) for document/table understanding: Qwen2-VL and Qwen2.5-VL (note per-size license ? which sizes are Apache-2.0 vs restricted), MiniCPM-V 2.6, InternVL2 / 2.5, Microsoft Phi-3.5-vision, Mistral Pixtral, Meta Llama-3.2-Vision (flag the Llama Community License restrictions), Moondream2, GOT-OCR2.0, olmOCR and other OCR-specialized VLMs.
- OCR-specialist / document-AI pipelines: PaddleOCR + PP-Structure (table recognition), docTR, EasyOCR, Tesseract (already used), Microsoft Table Transformer / TATR (check code AND PubTables-1M dataset/weights license), Donut, Surya (FLAG license), Marker (FLAG license), MinerU / PDF-Extract-Kit, RapidOCR.

- Table-structure recognition specifically (cell/grid reconstruction for borderless tables): TATR, PP-Structure, UniTable, etc.

Also assess: for SEMI-RULED / borderless bank tables, is a prompt-driven VLM (page image -> JSON transactions) more reliable than a classic OCR + table-structure pipeline? What are the accuracy/hallucination risks of VLMs on financial NUMBERS (transposed digits, dropped/duplicated rows), and how should output be VERIFIED? (muneem already machine-checks parses by balance reconciliation ? opening + sum(deposits) - sum(withdrawals) == closing, plus per-row running-balance walks and cross-checks against the bank's printed summary ? so any model's output can be automatically validated, which should inform the recommendation.)

Finally, briefly cover the best PAID/CLOUD fallbacks (clearly flagged approval-gated for sensitive data): AWS Textract (AnalyzeExpense / Tables), Google Document AI, Azure AI Document Intelligence, Reducto, LlamaParse, Anthropic/OpenAI vision ? note any self-hosting option and data-handling terms.

Deliver a RANKED shortlist of the most promising LOCAL options for muneem (NVIDIA GPU, Python, commercial-license-safe), each with license (code + weights), VRAM tier, maturity, strengths/weaknesses for dense bank-statement tables, and a concrete recommendation: which local model(s) to prototype for muneem's statement parsing and how they'd combine with the existing deterministic parsers + balance-reconciliation verification (e.g. VLM as a fallback/cross-check when deterministic parsing fails to reconcile). Cite sources and call out any license uncertainty explicitly.

Claim under review

"Surya's model WEIGHTS are NOT permissively licensed: they use a modified AI Pubs OpenRAIL-M license that is free only for research, personal use, and organizations under \$5M funding/revenue, with a commercial license required above that threshold. This fails muneem's requirement that model weights be MIT/BSD/Apache-2.0 for unrestricted commercial use."

****Source:**** <https://github.com/VikParuchuri/surya/blob/master/LICENSE> (primary)

****Supporting quote:**** "The model weights use a modified AI Pubs Open Rail-M license (free for research, personal use, and startups under \$5M funding/revenue)"

Checklist

1. Is the claim actually supported by the quote, or is it an overreach/misread?
2. WebSearch for contradicting evidence ? does any credible source dispute or heavily qualify this?
3. Is the source quality sufficient for the claim's strength? (extraordinary claims need primary sources)
4. Is the claim outdated? (check dates ? old claims about fast-moving fields are suspect)
5. Is this a marketing claim / press release / cherry-picked benchmark / forum speculation?

****refuted=true**** if: unsupported by quote / contradicted / low-quality source for strong claim / outdated / marketing fluff.

****refuted=false**** ONLY if: claim is well-supported, current, and source quality matches claim strength.

Default to refuted=true if uncertain.

Structured output only. Evidence MUST be specific.

2. assistant (2026-05-31T04:22:20.375Z)

I'll verify this claim about Surya's model weights license. Let me check both the primary source and look for any contradicting or updated information.

3. assistant (2026-05-31T04:22:24.150Z)

I have the tools I need. Let me verify this claim by checking the primary source and searching for any contradicting or updated evidence.

4. user (2026-05-31T04:22:28.433Z)

License Information for Surya

Based on the webpage content provided, the **code** is licensed under the Apache License 2.0.

However, **the webpage does not contain information about model weights licensing, commercial use restrictions, research use policies, funding thresholds, OpenRAIL-M, or AI Pubs licenses.** The displayed content shows only the Apache 2.0 license text for the code repository.

To find details about model weights licensing and usage restrictions, you would need to check other documentation in the Surya repository, such as a README file or separate model license documentation.

5. user (2026-05-31T04:22:31.029Z)

Apache License 2.0 Analysis

Based on the provided text, here's what the license addresses:

Code License vs Model Weights: The license makes no distinction between code and model weights?it applies uniformly to "the Work" in any form.

Commercial Use: Permitted. The license grants "no-charge, royalty-free" rights including "offer to sell, sell, import, and otherwise transfer the Work."

Research/Personal Use: Not specifically addressed. The license doesn't differentiate between use categories.

Startups: No special provisions. All entities are treated equally under the terms.

Funding/Revenue Thresholds: Not mentioned. The license contains no \$5M threshold or any funding-based restrictions.

Key Quote:

"Each Contributor hereby grants to You a perpetual, worldwide, non-exclusive, no-charge, royalty-free, irrevocable copyright license to reproduce...and distribute the Work" (Section 2).

The Apache 2.0 is a permissive license with minimal restrictions?it allows broad commercial and non-commercial use alike, provided users comply with attribution and modification notification requirements.

6. assistant (2026-05-31T04:22:34.932Z)

Interesting. The LICENSE file in the repo is pure Apache 2.0 ? it does NOT contain the model weights restriction language quoted in the claim. The claim's supporting quote ("The model weights use a modified AI Pubs Open Rail-M license...") does NOT appear in the LICENSE file. This is a critical finding. The model-weights restriction is documented in the README, not the LICENSE file. Let me verify where the actual license terms live and what they currently say.

7. user (2026-05-31T04:22:39.676Z)

Licensing Information for Surya

Code License:

"Code License-Apache--2.0" ? Surya's source code uses the Apache 2.0 license.

Model Weights License:

The model weights employ "a modified AI Pubs Open Rail-M license" with these provisions:

- **Permitted uses:** Research, personal projects, and startups with less than "\$5M funding/revenue"
- **Commercial licensing:** Beyond the startup threshold requires visiting Datalab's pricing page for broader commercial licensing arrangements

****Key Quote:****

"free for research, personal use, and startups under \$5M funding/revenue"

The documentation explicitly notes that those needing broader commercial licensing of the model weights should consult their pricing page for available options.

8. user (2026-05-31T04:22:42.733Z)

Web search results for query: "Surya OCR VikParuchuri model weights license GPL commercial 2025"

Links: [{"title": "GitHub - datalab-to/surya: OCR, layout analysis, reading order, table recognition in 90+ languages · GitHub", "url": "https://github.com/datalab-to/surya"}, {"title": "GitHub - CibiQuadance/surya-OCR: OCR and line detection in 90+ languages · GitHub", "url": "https://github.com/CibiQuadance/surya-OCR"}, {"title": "surya-ocr · PyPI", "url": "https://pypi.org/project/surya-ocr/0.1.2/"}, {"title": "Surya OCR | tasarim.ai", "url": "https://tasarim.ai/en/models/surya-ocr"}, {"title": "pitapo/surya · Hugging Face", "url": "https://huggingface.co/pitapo/surya"}, {"title": "surya/MODEL_LICENSE at master · datalab-to/surya", "url": "https://github.com/datalab-to/surya/blob/master/MODEL_LICENSE"}, {"title": "GitHub - cognitiveailab/surya-sci: OCR, layout analysis, reading order, line detection in 90+ languages · GitHub", "url": "https://github.com/cognitiveailab/surya-sci"}, {"title": "GitHub - kunthet/surya-text-line-detection-ocr: Accurate line-level text detection and recognition (OCR) in any language · GitHub", "url": "https://github.com/kunthet/surya-text-line-detection-ocr"}, {"title": "Datalab develops small AI models to power 'document intelligence' in the enterprise", "url": "https://www.forkable.io/p/datalab-develops-small-ai-models"}, {"title": "surya/LICENSE at master · datalab-to/surya", "url": "https://github.com/VikParuchuri/surya/blob/master/LICENSE"}]

Based on the search results, here's what I found about Surya OCR's licensing:

Code License

The Surya code is licensed under Apache 2.0.

Model Weights License

There appear to be some differences in the licensing information across sources:

****Current Status (2025):**** The model weights use a modified AI Pubs Open Rail-M license (free for research, personal use, and startups under \$5M funding/revenue).

****Commercial Licensing:**** For broader commercial licensing of the model weights, visit the pricing page.

GPL Considerations

Regarding GPL and commercial use: Most of the code is available under an open source GPL license, however the underlying model weights are licensed for non-commercial use only. While anyone is free to inspect, modify, and redistribute the code, they can't use the trained models in any commercial setting without a paid license.

Dual Licensing Option

To remove GPL license requirements or use the weights commercially over the revenue limit, you can contact surya@vikas.sh for dual licensing.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

9. assistant (2026-05-31T04:22:45.607Z)

I now have strong confirmation. The README explicitly contains the quoted language. The repo has

moved to `datalab-to/surya` and there's a dedicated `MODEL\_LICENSE` file. Let me verify the dedicated MODEL_LICENSE file to confirm the exact terms and check the threshold is precisely \$5M (revenue or funding).

10. user (2026-05-31T04:22:51.052Z)

AI PUBS OPEN RAIL-M LICENSE (MODIFIED)

****License Name:**** AI PUBS OPEN RAIL-M License (Modified), Version 0.1, March 2, 2023

****Funding/Revenue Thresholds:**** The license establishes two separate \$5,000,000 thresholds:

1. ****Gross Revenue:**** "for any purpose if You (your employer, or the entity you are affiliated with) generated more than two million US Dollars (\$5,000,000) in gross revenue in the prior year"

2. ****Equity/Debt Funding:**** "for any purpose if You (your employer, or the entity you are affiliated with) has raised more than two million US dollars (\$5,000,000) in total equity or debt funding"

****Research/Personal Use Exception:**** Yes. Both restrictions include carve-outs: "except where Your Use is limited to personal use or research purposes"

****Commercial Use Restrictions:**** Yes. Beyond the threshold restrictions, there is a broader prohibition: "for any purpose if You (your employer, or the entity you are affiliated with) provides or otherwise makes available any product or service that competes with any product or service offered by or made available by Licensors"

The license notes that "Commercial and broader use licenses may be available from Licensors" at their specified URL.

11. assistant (2026-05-31T04:23:03.885Z)

I have now fully verified the claim from the primary source ? the dedicated `MODEL\_LICENSE` file in the official `datalab-to/surya` repo.

Let me assess the claim against the checklist:

****1. Is the claim supported by the quote?**** Yes, and verified beyond the quote. The `MODEL\_LICENSE` file confirms it is the "AI PUBS OPEN RAIL-M License (Modified)" with a \$5,000,000 threshold on both gross revenue AND equity/debt funding, with carve-outs only for personal/research use. (Minor note: the license prose says "two million US Dollars (\$5,000,000)" ? a typo in the original where the words say "two million" but the figure says \$5M; the operative figure is \$5M, consistent with the claim and README.)

****2. Contradicting evidence?**** None found. Multiple independent sources (README, search results, MODEL_LICENSE) all corroborate. One search result noted the code was historically "GPL" before relicensing to Apache 2.0 ? but that strengthens, not weakens, the "not permissive" concern.

****3. Source quality?**** Excellent ? primary source (the license file itself in the official repo).

****4. Outdated?**** Current as of 2025-2026. The repo moved to `datalab-to/surya` (Datalab is the commercial entity), and the model-weights restriction persists. If anything, the commercial-licensing posture has hardened.

****5. Marketing/speculation?**** No ? it is the actual binding license text.

****Important nuance on the claim's wording:**** The claim says weights "fail muneem's requirement that model weights be MIT/BSD/Apache-2.0." This is correct. The AI Pubs OpenRAIL-M (Modified) is a restrictive RAIL license, not permissive, and the research/CLAUDE.md constraint demands MIT/BSD/Apache-2.0/ISC/HPND. The claim also correctly characterizes it as "free only for research, personal use, and organizations under \$5M." For muneem specifically (single-user personal tool), the personal-use carve-out would actually permit Avi's own use ? but the claim

is about the **commercial-license-safety requirement** in the abstract, and on that point the claim is accurate: the weights are NOT permissively licensed and DO require a commercial license above the threshold.

The claim is well-supported, current, sourced to primary documents, and accurate. It is NOT refuted.

12. user (2026-05-31T04:23:13.311Z)

Structured output provided successfully

13. assistant (2026-05-31T04:23:18.655Z)

The claim is verified and NOT refuted. It is well-supported by the primary source (the MODEL_LICENSE file in the official datalab-to/surya repository), corroborated by the README, current as of 2025-2026, and accurate in its core assertion that Surya's model weights use a restrictive AI Pubs OpenRAIL-M (Modified) license ? not a permissive MIT/BSD/Apache-2.0 license ? and therefore fail muneem's commercial-license-safety requirement.

One sourcing nuance worth flagging for the synthesis: the claim's cited supporting quote actually originates from the README, not the LICENSE file. The bare LICENSE file is pure Apache-2.0 (that covers only the code). The binding weights restriction lives in the separate MODEL_LICENSE file. This doesn't refute the claim ? it confirms it via a better source ? but the citation should point at MODEL_LICENSE/README rather than LICENSE.